UNIHUB

Platform Feature Inventory & Code Audit

Multi-tenant university events app (Flutter) — project *uni_events_app* v1.0.0+1
Report date: 20 October 2026

Metric	Value
Framework	Flutter (Dart ≥3.0), Material 3, Provider state management
Source size	~5,700 lines in lib/ (18 Dart files + 2 ARB files)
Platforms scaffolded	Android, iOS, Web, Windows, macOS, Linux
Backend	**None** — all data is in-memory mock data; Firebase is commented out
Version control	**Not a git repository**
flutter analyze	76 info-level lints, 0 errors, 0 warnings
flutter test	1 test, **1 failing**

1. Executive Summary

UNIHUB is a polished front-end prototype for a university events platform. It has two audiences — students (discover events, register, hold QR tickets, leave reviews) and planners (publish events, scan tickets, view analytics) — plus a per-university (tenant) theming system and English/German localization. The UI layer is well developed. Everything behind it is simulated: there is no authentication, database, API, persistence, or real multi-tenant isolation, and the planner dashboard shows largely hard-coded figures.

Bottom line: the app is a convincing demo, not yet a platform. The biggest steps to a complete product are (1) a real backend with authentication and tenant-scoped data, (2) a secure, server-validated ticketing flow, (3) real analytics replacing mock numbers, and (4) closing the bugs in section 4 — several of which would block an App Store/Play release.

2. Architecture Overview

Layer	Files	Notes
Entry / App shell	main.dart, features/shell/shell_screen.dart	MaterialApp, theme + locale from state; role-based bottom nav (<600px) / NavigationRail (≥600px); profile switcher in header
State	core/auth/user_role.dart (AppState)	Single ChangeNotifier holding role, locale, tenant, theme, wallpaper, registrations, tickets, scans, reviews, mock profiles
Models	core/models/event_model.dart, review_model.dart	Event (immutable, copyWith), Review
Data	core/services/event_repository.dart	In-memory list of 5 seeded events; add only
Student UI	features/student/*	Home, Details, Tickets, Review, Profile
Planner UI	features/planner/*	Dashboard, Create Event, QR Scanner
Theme / i18n	app/config/theme.dart, l10n/*	Light/dark themes seeded by tenant colour; ~35 translated strings (en, de)

3. Current Feature Inventory

Status key: **Working** = functional within the session; **Mock** = UI exists but data/behaviour is simulated; **Stub** = visible control that does nothing.

3.1 Platform-wide

Feature	Status	Details
Role-based navigation	Working	Student (Home, Tickets, Profile) and Planner (Dashboard, Create, Scanner) shells; responsive bottom bar / side rail
Profile switcher	Mock	Three hard-coded personas: student (tum.de), university organizer (tum.de), commercial planner (public). Switching changes role, planner type and tenant
Multi-tenant theming	Working	Brand colour per university: TUM blue, LMU green, UHH red; applied to theme, chips, badges, borders
Localization	Working (partial)	English + German via ARB; language switch in Profile. Many strings are still hard-coded English (see bugs)
Theme mode	Working	System / Light / Dark selector
Wallpapers	Working	Minimal Velvet, Frosted Geometry, Cyber Aurora backgrounds for students (planners forced to Minimal Velvet)

3.2 Student

Feature	Status	Details
Event discovery feed	Working	Two tabs: Campus Hub (official university events) and Public Events; responsive 1/2/3-column grid; image cards with date badge
Category filter	Working	All, Music, Tech, Sports, Arts chips; multi-category events match any
Event details	Working	Hero banner, official/public badge, category tags, date, location, rating, description, image gallery (category-based stock photos)
Registration / RSVP	Mock	One-tap register from card or details page; generates ticket ID; no capacity, cancel or waitlist
My Tickets	Mock	Ticket cards with QR code (qr_flutter), perforated-ticket design, seat info, CHECKED IN state once scanned
Reviews	Working	1–5 star + comment. QR events: only after check-in (verified). Non-QR events: self-claimed attendance (unverified)
Profile & settings	Partial	Avatar/initials, email, role badge, language, tenant, theme, wallpaper. Push-notification switch, Privacy & Data row and Sign Out are non-functional

3.3 Planner

Feature	Status	Details
Dashboard KPIs	Mock	Total RSVPs, check-ins, attendance rate (base numbers 320/240 hard-coded + live session counts), active events count
Attendance by event	Mock	Three hard-coded events with fixed progress bars
Scanner logs	Mock	Three hard-coded log lines
Feedback inbox	Working	Verified reviews (public ratings) separated from unverified (private feedback) with disclaimer
Create event	Working	Title, up to 3 categories, location, date/time picker, capacity slider, QR toggle, reviews toggle, description, live student-card preview; official status inferred from planner type
QR ticket scanner	Working / Mock	Live camera via mobile_scanner, JSON ticket parsing, duplicate-scan prevention, Access Granted/Denied sheet, torch toggle. A ‘Mock Scan Ticket’ button simulates scans

4. Bugs & Issues Found

Severity: **High** = breaks core function or blocks release; **Med** = wrong behaviour or product risk; **Low** = polish/maintenance. All items below were confirmed by reading the code; items marked (analyzer) or (test) were also reproduced by tooling.

#	Sev	Issue	Where	Impact / fix
1	High	Smoke test fails: expects exactly one school icon but finds two (header logo + Campus Hub tab icon). (test)	test/widget_test.dart:25	Only test in the project is red. Use a key/text finder; also stub Image.network (HTTP returns 400 in tests).
2	High	Camera permission missing: no NSCameraUsageDescription in iOS Info.plist; no CAMERA permission in the app's Android manifest (plugin may merge it — verify).	ios/Runner/Info.plist, android/.../AndroidManifest.xml	iOS terminates the app when the scanner opens. Add usage string and permission.
3	High	Ticket QR is unsigned plain JSON (ticketId, userName). Anyone can forge a payload; validity is checked only against in-memory ticket IDs.	tickets_screen.dart:93, user_role.dart validateAndScanTicket	Needs server-issued signed/rotating tokens and server-side validation.
4	High	Scanner accepts any ticket for any event and any planner. A commercial planner can check in university tickets; no event/organizer binding.	user_role.dart validateAndScanTicket	Scope scans to the planner's own events; verify eventId.
5	High	No real auth: Sign Out is a snackbar; users are 3 hard-coded personas; QR payload hard-codes userId student-1 / Alex Smith regardless of active profile.	profile_screen.dart:177, tickets_screen.dart:93	Implement auth (Firebase Auth planned) and real user IDs; verify university e-mail domains.
6	High	All state is in memory: restart loses registrations, tickets, scans, reviews, created events. Dashboard counters are partly fabricated (320 + n, 240 + n).	user_role.dart, event_repository.dart	Persist to a backend; compute analytics from real data.
7	Med	Tenant isolation not enforced: Campus Hub lists official events from every university; the tenant dropdown lets any student switch to any university.	home_screen.dart:35, profile_screen.dart:52	Filter by verified tenant; derive tenant from account.
8	Med	Capacity collected in Create Event but discarded — Event model has no capacity, so registration is never limited and the dashboard cannot show real fill rates.	create_event_screen.dart:21, event_model.dart	Add capacity to model; enforce on register.
9	Med	Dashboard shows all reviews from all events/organizers; if an event is missing it silently falls back to events.first (wrong title).	dashboard_screen.dart:137,180	Filter by the planner's events; handle missing event explicitly.
10	Med	Event rating shows a fake 4.5 when there are no reviews; includes unverified reviews; labelled ‘Organizer Rating’ though it is per-event; star display uses floor().	details_screen.dart:24,170	Show ‘No ratings yet’; average verified only.
11	Med	Review integrity: duplicates allowed per user/event; non-QR events can be reviewed before they happen; ‘verified’ flag is derived from the event setting, not from an actual scan.	review_screen.dart:171, tickets_screen.dart:318	One review per ticket; require event date passed; set verified from scan record.
12	Med	Scanner camera is built inside an IndexedStack, so it starts on login as planner and keeps running on Dashboard/Create tabs; permission prompt appears unexpectedly.	shell_screen.dart:117, qr_scanner_screen.dart	Start/stop controller only when the Scanner tab is active.

#	Sev	Issue	Where	Impact / fix
13	Med	Invented ticket seating (hash-based section/row/seat) is shown for all events including general admission; duplicates are possible; String.hashCode is not guaranteed stable across runs.	tickets_screen.dart:96-99	Remove, or implement real seat assignment.
14	Med	Two stacked headers: shell AppBar plus each screen's own AppBar (nested Scaffolds) waste vertical space.	shell_screen.dart:51, home_screen.dart:41	Single app bar owner.
15	Med	Details page: Image.network has no errorBuilder (offline/broken URL shows an error); every user-created event falls back to the same stock concert photo; no image upload.	details_screen.dart:59	Add errorBuilder, image upload.
16	Med	No past-event handling: events never expire; registration allowed for past events; no cancel-registration; every event says ‘FREE ADMISSION’.	event_repository.dart, details_screen.dart:358	Add upcoming/past split, cancel, pricing.
17	Low	Widespread hard-coded English strings despite DE localization (~100 across Home, Details, Tickets, Profile, Create, Dashboard, Scanner), plus raw category keys and d.m.yyyy dates not locale formatted.	all screens	Move to ARB; use intl DateFormat.
18	Low	Create-event form pre-filled with demo values (‘TUM Chess Championship 2026’); titles not trimmed so spaces pass validation; past times selectable today; BuildContext used across async gap. (analyzer)	create_event_screen.dart:16,115,163	Blank defaults, trim, validate future time, check mounted.
19	Low	Dead or desynced code: toggleRole/setRole bypass the active-profile index; commercial planner inherits a university tenant name/colour in the header.	user_role.dart:130-140	Derive role from profile only.
20	Low	~70 deprecated withOpacity / activeColor calls (analyzer); tiny 8–9 px fonts hurt accessibility; EventRepository sorts its list inside a getter on every read.	project-wide	Migrate to withValues; raise minimum font size.
21	Low	Project hygiene: not under git; package id com.example.uni_events_app; boilerplate README; flutter_01.log committed to folder; firebase deps commented out.	project root	git init, real bundle ID, README, .gitignore.

5. Suggested Roadmap to a Complete Platform

Phase	Goal	Key work
0 — Foundations (days)	Make the repo safe to build on	git init + remote; fix failing test; add camera permissions; real app ID; README; CI running analyze + test
1 — Backend & Auth	Replace mocks with real data	Firebase (or custom API): auth with university e-mail verification, Firestore/Postgres for events, tickets, reviews, users; repository interfaces behind AppState; offline cache
2 — Tenancy & roles	Real multi-tenant platform	Tenant model + domain verification; roles (student, organizer, tenant admin, platform admin); organizer approval workflow; security rules
3 — Secure ticketing	Trustworthy check-in	Signed rotating QR tokens, server validation, per-event scan permissions, offline scan queue, capacity & waitlist, cancellations, wallet pass
4 — Real analytics	Planner value	Per-event RSVPs/check-ins/ratings from live data, exports (CSV), scan log history, feedback moderation and organizer replies

Phase	Goal	Key work
5 — Engagement	Growth	Push notifications & reminders, calendar sync, search, favourites, follow organizers/clubs, maps, image upload, sharing / deep links
6 — Monetisation & ops	Commercial planners	Paid tickets (Stripe), organizer billing, moderation/reporting, GDPR tooling (data export/delete), admin console
7 — Quality & release	Ship	Full i18n pass, accessibility audit, widget/integration tests, crash reporting, store listings

Method: manual review of every Dart source, ARB, theme, test and platform config file; flutter analyze and flutter test were executed. The app was not run on a device, so layout overflow and runtime camera behaviour are unverified.